

Person 3 Expanded Analysis Report

Ma two-population beta-lactam/Bla inhibitor model; Person 3 analysis

This report presents the exhaustive Person 3 analysis. The primer below defines the model before presenting the analysis, so the plots can be interpreted mechanistically rather than as purely mathematical parameter sweeps.

Model equations and mechanistic interpretation

The model is dimensionless. τ is scaled time; n_s and n_r are susceptible and resistant cell abundances; s is the shared nutrient/resource pool; a is active antibiotic; and b is extracellular free beta-lactamase (Bla). The equations below are the Ma paper/supplement base model used for this local analysis.

$$dn_s/d\tau = (g - \ell)n_s$$

$$dn_r/d\tau = (\alpha g - \beta \ell)n_r$$

$$ds/d\tau = (\xi \ell - g)n_s + (\xi \beta \ell - \alpha g)n_r$$

$$da/d\tau = -\kappa_\beta ba - \phi n_r a - d_a a$$

$$db/d\tau = \beta \ell n_r - d_\beta b$$

$$g = s/(1+s)$$

$$\ell = \gamma [a^{h_a}/(1+a^{h_a})] g$$

$$\iota = i^{h_i}/(1+i^{h_i})$$

$$\beta = \beta_{\min} + c(1-\beta_{\min})\iota$$

$$\phi = \phi_{\max}(1-c\iota)$$

Mechanistic reading: g is nutrient-limited bacterial growth; ℓ is antibiotic-mediated lysis; ι is the fractional inhibitor activity; β is the resistant-cell lysis factor after inhibitor action; and ϕ is the rate at which living resistant cells degrade antibiotic. Free Bla is produced when resistant cells lyse and is removed at the inhibitor-dependent rate $d_\beta \iota$. Thus, inhibitor activity simultaneously weakens private resistance protection through β and public antibiotic degradation through ϕ .

State/control	Where it appears	Mechanistic real-world equivalent
n_s	First ODE, nutrient and lysis terms	Number or density of antibiotic-susceptible bacteria; the population that experiences the full lysis rate ℓ .
n_r	Second, third, fourth, and fifth ODEs	Number or density of resistant bacteria; the population carrying Bla/resistance and paying the growth burden α .
s	g and the nutrient-balance equation	Shared nutrient or limiting resource available to both populations; lysis can recycle part of it through ξ .
a	ℓ and antibiotic-loss equation	Active antibiotic concentration or effective antibiotic exposure in the environment.
b	Antibiotic-loss and Bla-balance equations	Free extracellular Bla enzyme; a public-good pool that degrades antibiotic but is inactivated by the inhibitor.
τ	All differential equations	Dimensionless time; map to experimental time only after specifying the model time scale.

Parameter	Equation role	Mechanistic real-world equivalent	Effect of increasing it
α	αg in resistant growth; αg in nutrient use	Fitness of resistant bacteria relative to susceptible bacteria when antibiotic pressure is absent; $\alpha < 1$ represents a resistance cost.	Raises resistant growth and nutrient consumption; usually makes resistant enrichment more likely.
$\beta_{\min}$	$\beta \ell$ without inhibitor	Baseline resistant-cell lysis sensitivity. $1 - \beta_{\min}$ is the private protection supplied by resistance/Bla inside or near the cell.	Raises resistant lysis at low inhibitor activity; reduces the private benefit of resistance.
ξ	$\xi \ell$ and $\xi \beta \ell$ in $ds/d\tau$	Fraction of lysed biomass converted back into shared usable nutrient; a proxy for recycling efficiency.	Increases resource regeneration after lysis and can support later regrowth.

Parameter	Equation role	Mechanistic real-world equivalent	Effect of increasing it
d_a	$-d_a a$	Background antibiotic clearance or decay independent of Bla and resistant cells; e.g., chemical degradation, dilution, or washout.	Shortens antibiotic persistence and generally reduces treatment pressure duration.
k_β	$-k_\beta b a$	Catalytic efficiency of free Bla in degrading active antibiotic; the strength of extracellular public protection.	Makes a given free-Bla concentration remove antibiotic faster.
γ	$\ell = \gamma \cdot \text{Hill}(a) \cdot g$	Maximum antibiotic-induced lysis strength relative to nutrient-limited growth; an antibiotic potency/lysis-scale parameter.	Increases lysis at a given antibiotic exposure and usually deepens the initial population bottleneck.
h_a	Hill exponent in ℓ	Steepness of the antibiotic concentration-response curve; how switch-like lysis is around the antibiotic transition.	Makes the antibiotic response more threshold-like, with effects depending on where a lies relative to the transition.
i	Hill exponent input for ι	Dimensionless inhibitor dose, scaled by the inhibitor half-maximal effective concentration (IC_{50} -like reference).	Increases inhibitor activity and therefore strengthens inhibitor effects when h_i is fixed.
h_i	Hill exponent in ι	Steepness/cooperativity of inhibitor action; how sharply inhibitor activity changes around $i=1$.	Increases ι for $i>1$ but decreases ι for $i<1$, so it is not globally equivalent to a stronger dose.
d_β	$-d_\beta t b$	Scale of inhibitor-dependent free-Bla inactivation/turnover; a proxy for enzyme neutralization or loss rate.	Shortens free-Bla persistence and reduces the duration of extracellular antibiotic protection.
$\phi_{\max}$	$\phi n_r a$	Maximum antibiotic degradation rate by living resistant cells; the cell-associated/public degradation pathway.	Makes resistant cells remove antibiotic more effectively when inhibitor activity is low.
c	β and ϕ inhibitor coupling	Fractional efficacy of inhibitor action on the resistance/protection pathway; it couples intracellular/private and public effects in this model.	Makes inhibitor activity more strongly increase β and decrease ϕ .
ι	β , ϕ , and $d_\beta t$	Effective fraction of inhibitor target activity, from zero to one; the direct pharmacodynamic variable generated by i and h_i .	Increasing ι weakens private protection, weakens public degradation, and increases free-Bla turnover.

Initial conditions are $n_s(0)=n_r(0)=0.2$, $s(0)=4$, $b(0)=0$, and $a(0)=a_0$. The analysis uses $a_0=2.15$, 10, and 100 as antibiotic anchors. The inhibitor dose i and its response steepness h_i are Person 3's main controls; the other parameters are frozen at the shared baseline unless a coordination/robustness run is explicitly identified.

Executive summary

The expanded atlas contains 3,159 simulations across three antibiotic anchors and three inhibitor parameter planes. $a_0=2.15$: median $\Delta f_R=-0.0094$, range [-0.0928, 0.0085], resistant-enriched fraction=0.36, susceptible-enriched fraction=0.63, recovered fraction=0.36 $a_0=10$: median $\Delta f_R=0.0846$, range [-0.4191, 0.4350], resistant-enriched fraction=0.62, susceptible-enriched fraction=0.38, recovered fraction=0.38 $a_0=100$: median $\Delta f_R=0.1884$, range [-0.4934, 0.4998], resistant-enriched fraction=0.62, susceptible-enriched fraction=0.38, recovered fraction=0.00

The primary interpretation is selection-relative: the change from the initial resistant fraction is reported alongside final composition. The inhibitor activity ι changes β and ϕ simultaneously in the full model, and d_b controls the duration of free-Bla protection through the effective rate $d_{\beta t}$. Therefore, matched inhibitor activity does not imply matched trajectories when d_b differs.

Methods and computational contract

The equations are integrated with LSODA using a log transform for n_s and n_r to prevent high-dose underflow from corrupting the composition signal. The environmental states remain in their original coordinates and are clipped to nonnegative values for reporting. The baseline is $\alpha=0.95$, $\beta_{\min}=0.90$, $\xi=0.80$, $d_a=0.02$, $k_\beta=0.35$, $\gamma=1.38$, $h_a=3$, $h_i=2$, $d_b=1$, $\phi_{\max}=1$, and $c=0.70$. Initial conditions are $n_s=n_r=0.2$, $s=4$, $b=0$, with antibiotic anchors $a_0=2.15$, 10, and 100. The full configuration is in `results/tables/analysis_config.json`.

The atlas uses i from 0.1 to 10 on a logarithmic grid, h_i from 1 to 5, and d_b from 1 to 10. A run is classified as neutral when $|\Delta f_R| \leq 0.001$. Recovery means n_s reaches 90 percent of the no-treatment reference endpoint and remains there in the final trajectory segment. Peak loss is one minus the minimum total live biomass divided by the initial total biomass.

Atlas results

a0	runs	Delta f_R median	Delta f_R range	R-enriched	S-enriched	recovered
2.15	1053	-0.0094	[-0.093, 0.009]	0.36	0.63	0.36
10	1053	0.0846	[-0.419, 0.435]	0.62	0.38	0.38
100	1053	0.1884	[-0.493, 0.500]	0.62	0.38	0.00

The corresponding response-atlas figures show whether the primary selection effect is accompanied by a biomass bottleneck, a change in free-Bla exposure, or loss of susceptible recovery. Companion maps should be read jointly: final f_R alone can be unfavorable even when total biomass is recovered, and a high f_R can also arise because the initial f_R was high.

Person 3 pharmacodynamic pathway

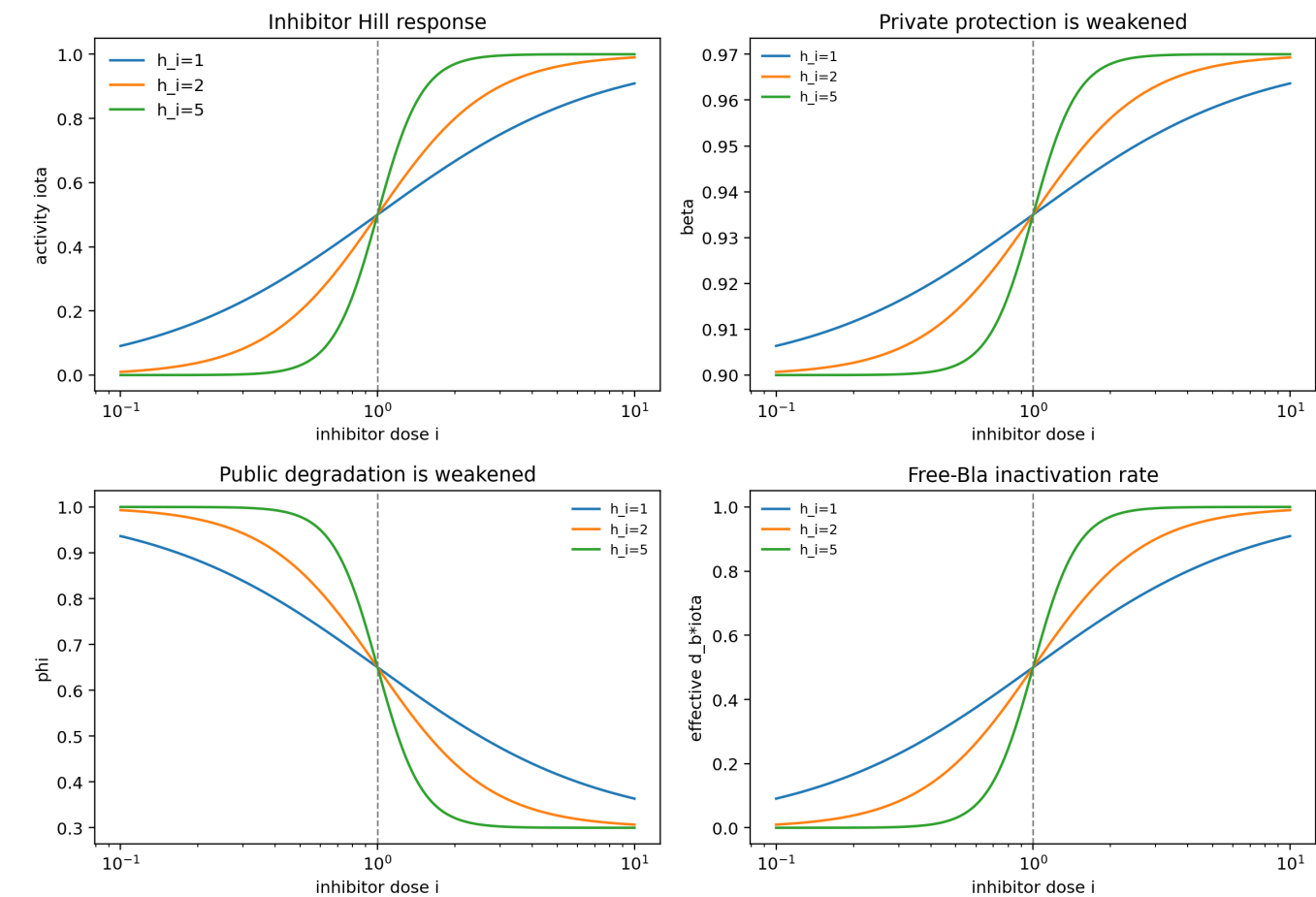


Figure 1. Effective inhibitor pathway: iota, beta, phi, and d_b iota.

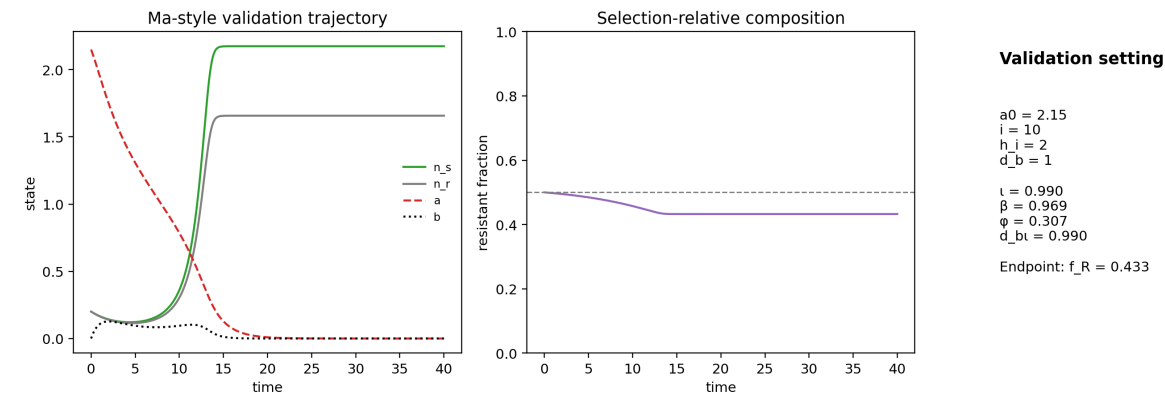


Figure 2. Representative Ma-style trajectory and resistant-fraction change.

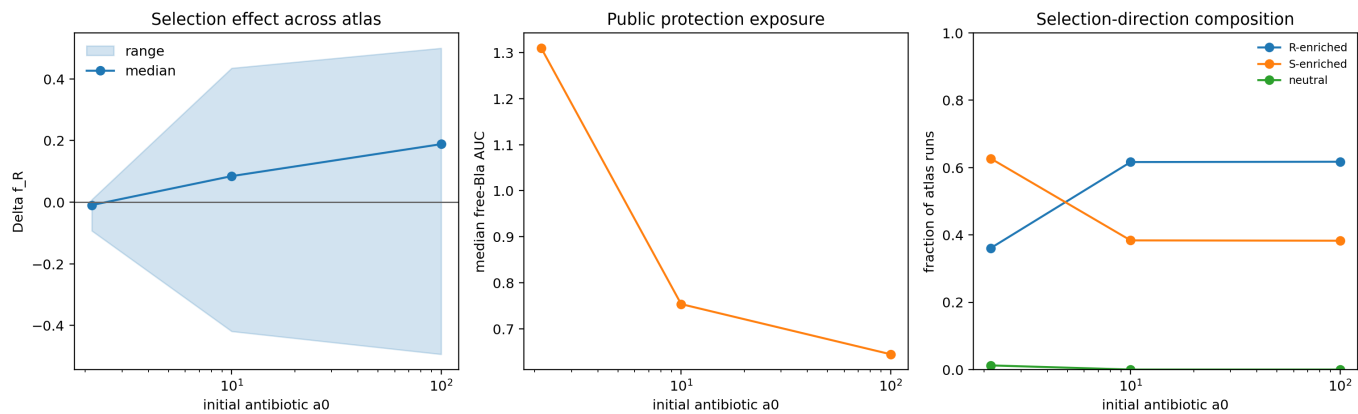


Figure 3. Cross-anchor comparison of selection and exposure.

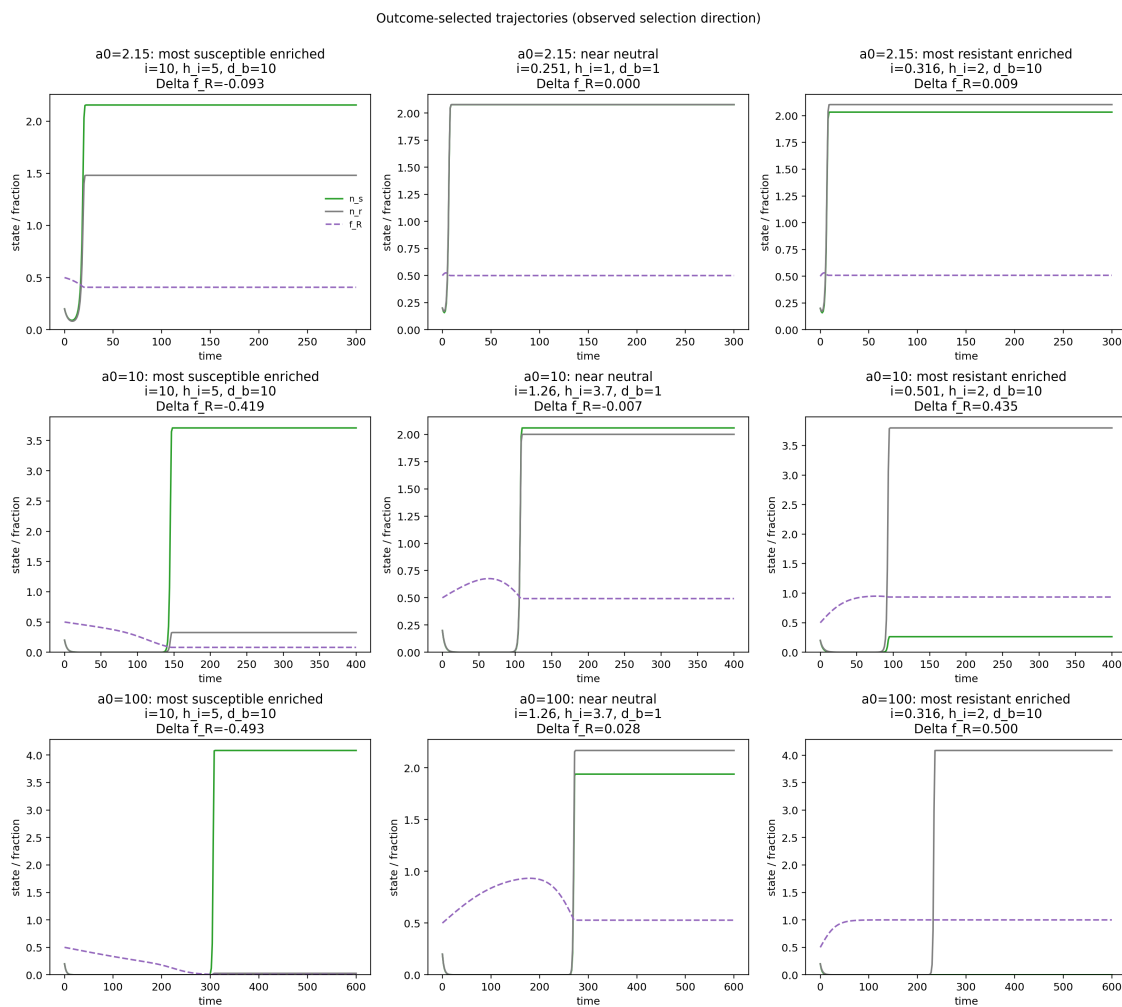


Figure 4. Algorithmically selected trajectories for susceptible-enriched, near-neutral, and resistant-enriched outcomes.

Initial composition

At fixed initial total biomass, changing $f_R(0)$ generally shifts the final composition as expected from the initial condition, but the slope and recovery behavior depend on the inhibitor regime. The selection-relative curves are the relevant test: if Δf_R keeps the same sign over the initial-composition grid, the inhibitor setting has a robust direction of selection; if it crosses zero, the initial mixture is part of the endpoint mechanism.

Fixed-total-biomass initial-composition analysis ($a_0=2.15$)

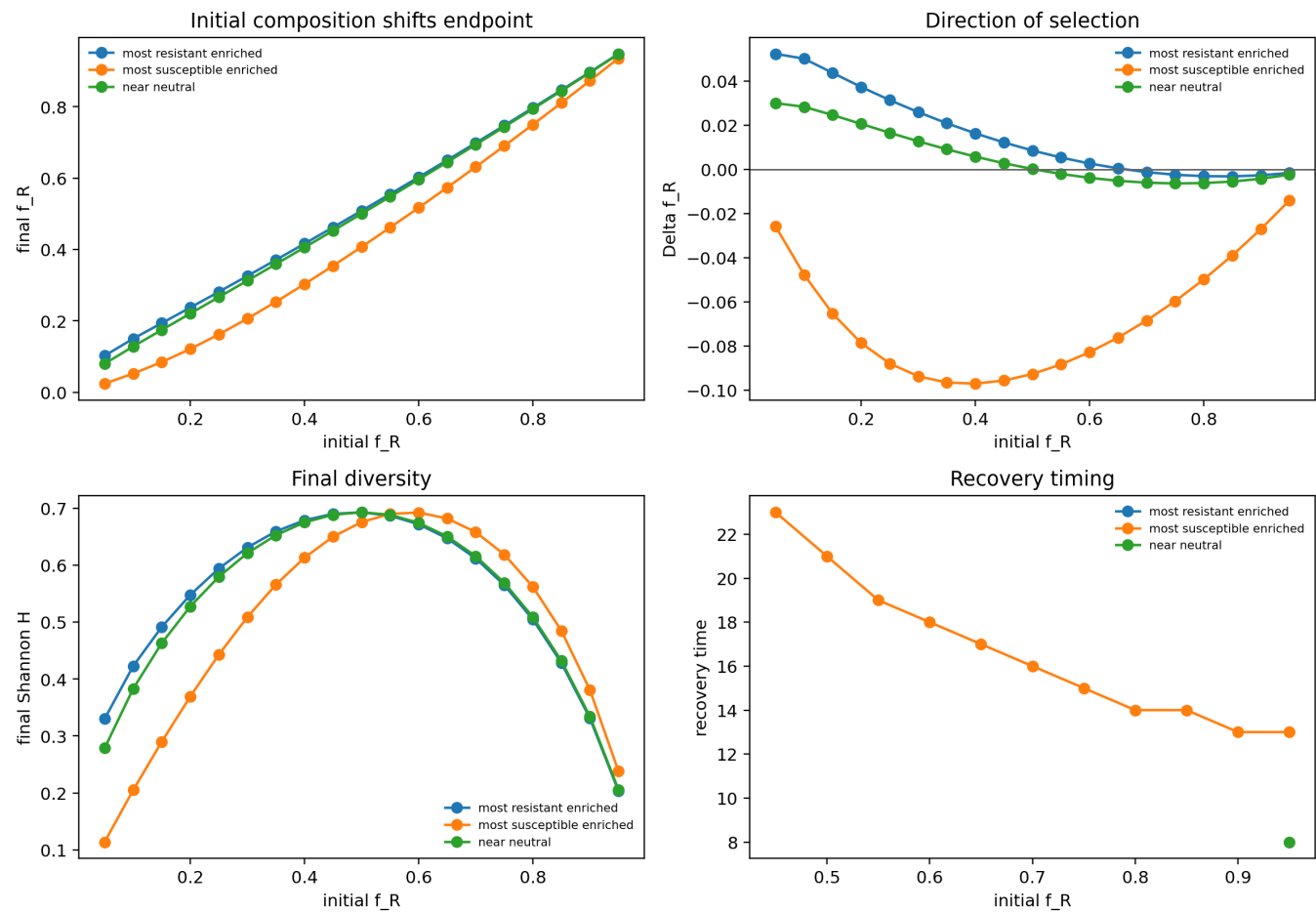


Figure 5. Initial resistant fraction versus endpoint composition, selection change, diversity, and recovery.

Mechanistic ablations

The ablations isolate the three routes by which inhibitor activity enters the model. Private-protection-only changes β ; public-degradation-only changes ϕ ; free-Bla-turnover-only changes $d_{B\text{iota}}$. Comparing these with the full pathway shows which mechanism explains the endpoint and which primarily changes the transient antibiotic/Bla exposure.

Mechanistic ablations at selected regimes

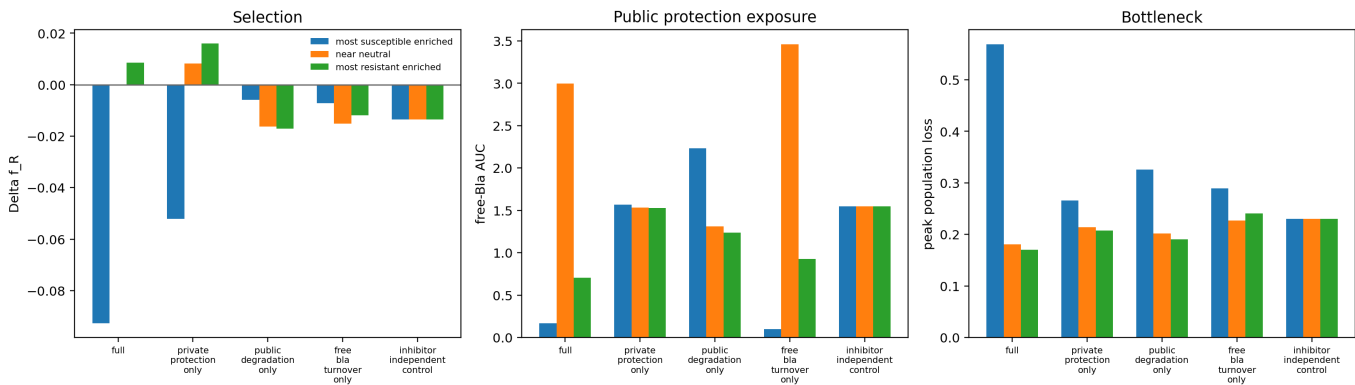


Figure 6. Mechanistic ablations across the selected low, neutral, and high selection regimes.

regime	mode	Delta f_R	free-Bla AUC	peak loss
most susceptible enriched	full	-0.0928	0.166	0.569
most susceptible enriched	private protection only	-0.0521	1.57	0.266
most susceptible enriched	public degradation only	-0.0059	2.23	0.326
most susceptible enriched	free bla turnover only	-0.0072	0.0968	0.289
most susceptible enriched	inhibitor independent control	-0.0135	1.55	0.230
near neutral	full	0.0001	3	0.181
near neutral	private protection only	0.0082	1.53	0.214
near neutral	public degradation only	-0.0162	1.31	0.202
near neutral	free bla turnover only	-0.0151	3.46	0.227
near neutral	inhibitor independent control	-0.0135	1.55	0.230
most resistant enriched	full	0.0085	0.703	0.171
most resistant enriched	private protection only	0.0160	1.53	0.207
most resistant enriched	public degradation only	-0.0170	1.24	0.191
most resistant enriched	free bla turnover only	-0.0119	0.928	0.241
most resistant enriched	inhibitor independent control	-0.0135	1.55	0.230

Ablation heatmaps across inhibitor settings

Each heatmap varies inhibitor dose i and Hill steepness h_i at $a_0=2.15$, with d_b fixed at its baseline. Figure 7 shows Δf_R within each simulation, measured from its own initial composition ($f_R(0)=0.5$). Figure 8 shows $\Delta\Delta f_R$: each mode's Δf_R minus the inhibitor-independent control's Δf_R at the same grid point. Thus the second set directly compares the ablation effect against the control. The full pathway and control are included as reference maps.

Ablation response maps at $a_0=2.15$
Within-run selection change, Δf_R

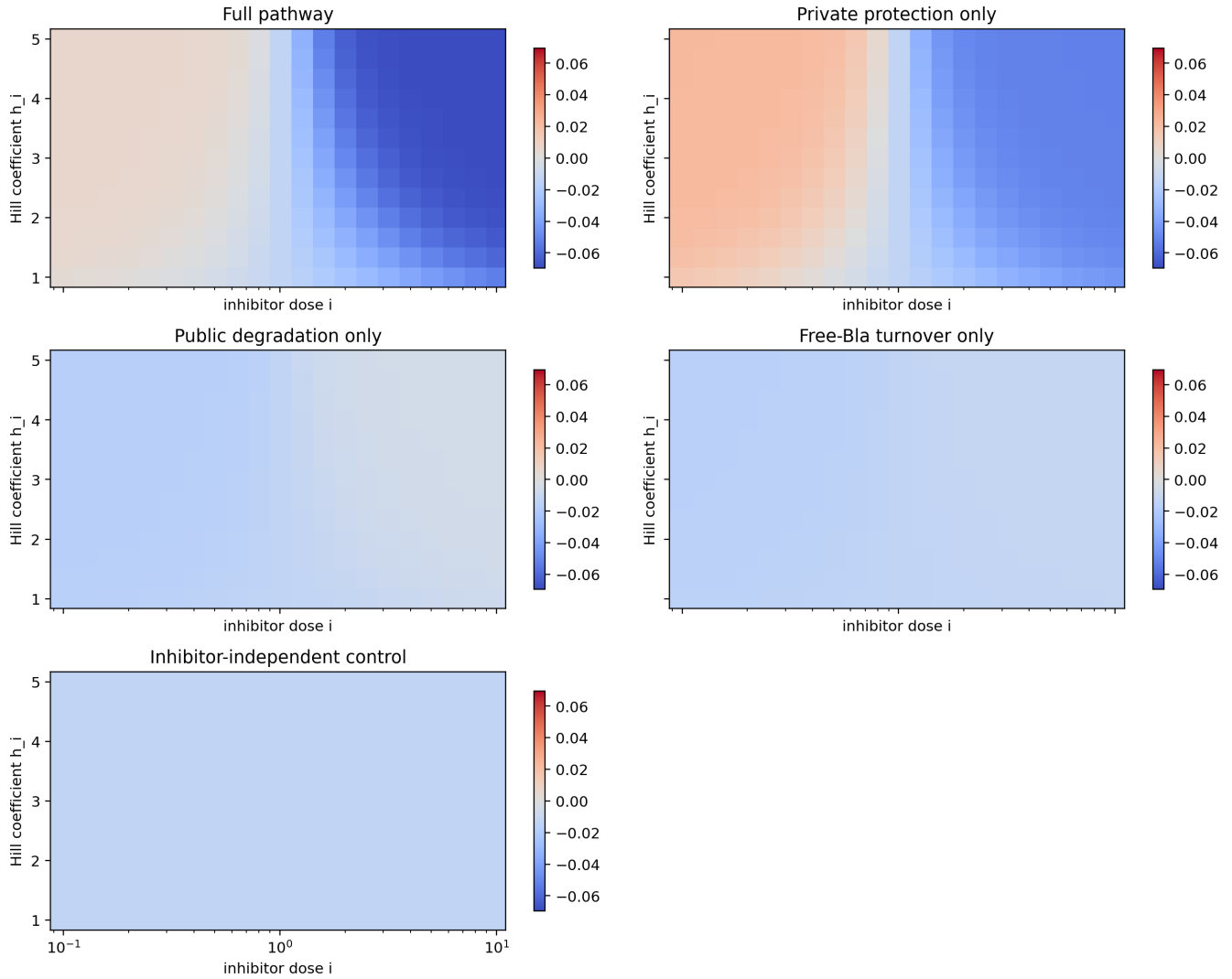


Figure 7. Selection change within each ablation setting.

Ablation response maps at $a_0=2.15$
Ablation effect relative to control, $\Delta\Delta f_R$

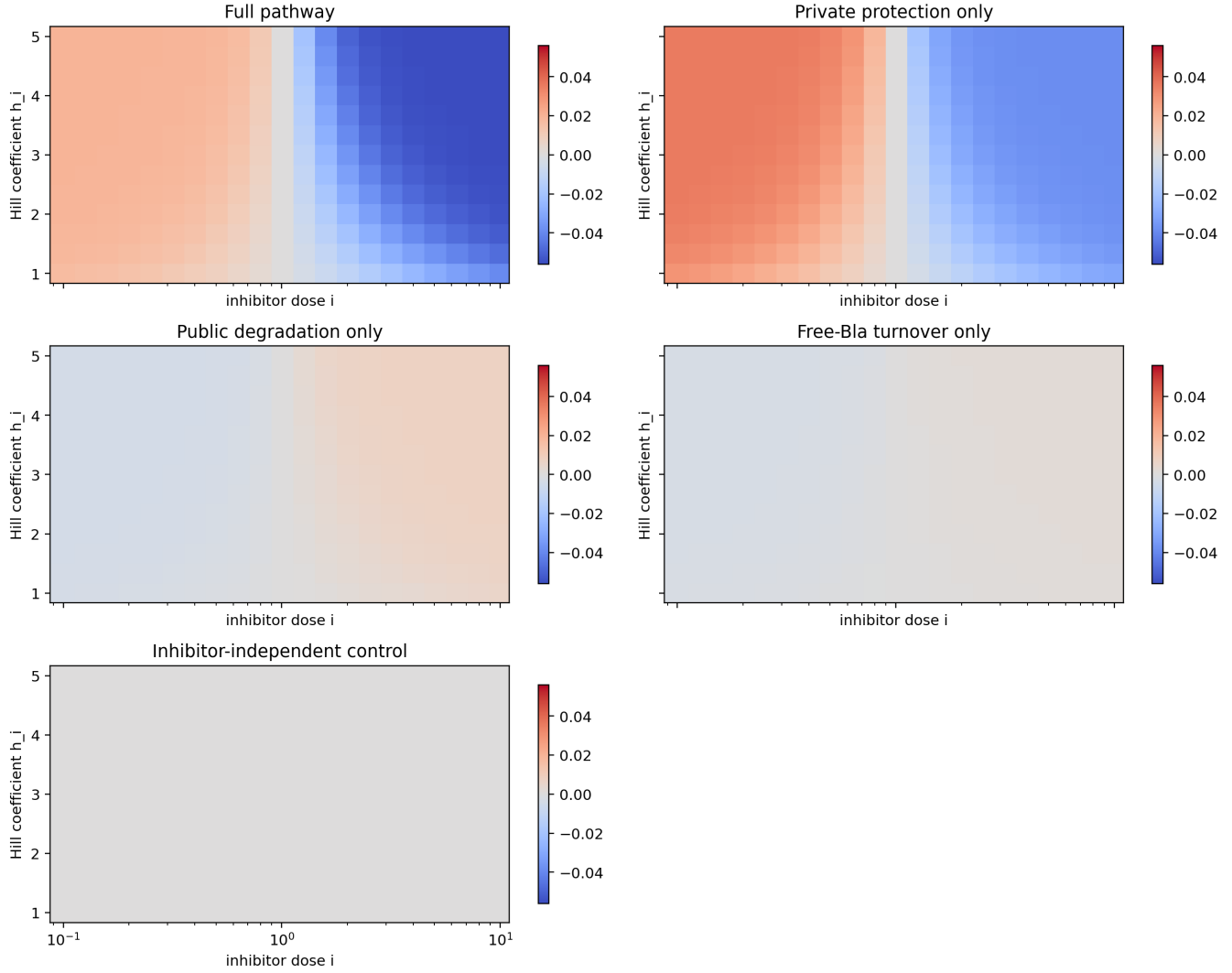


Figure 8. Difference in selection change relative to the matched inhibitor-independent control. Positive values mean more resistant enrichment (or less susceptible enrichment) than the control.

Robustness and identifiability

The robustness run samples the non-Person 3 parameters over the supplementary ranges. It is a coordination analysis: it tests whether the Person 3 interpretation survives plausible uncertainty, but it does not replace the group's agreed baseline. Rank correlations and the full sampled table are saved under results/tables.

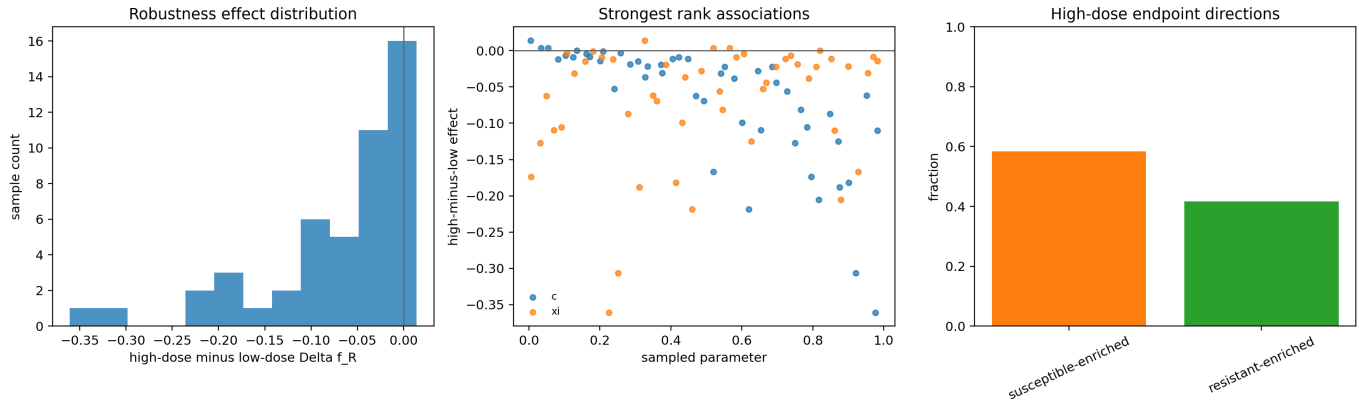
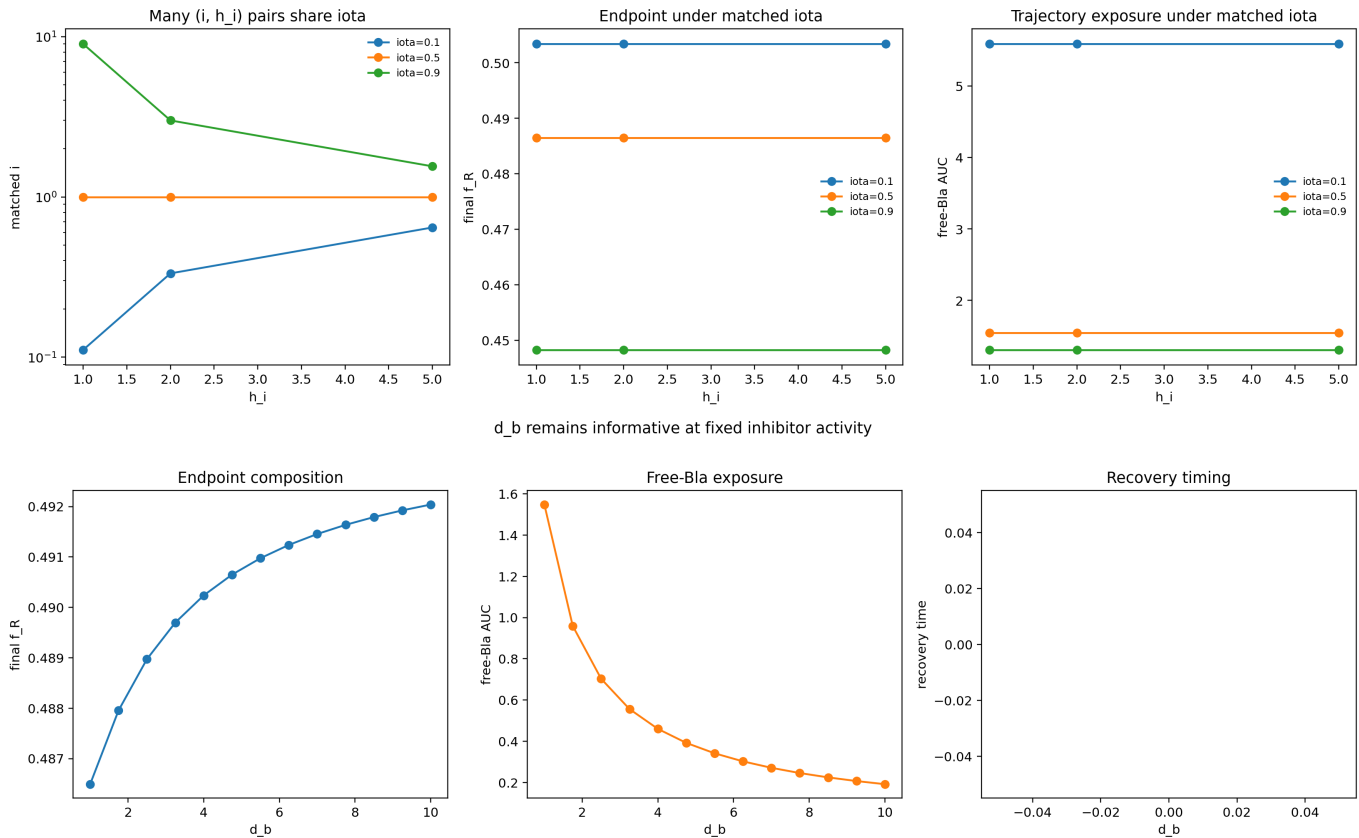


Figure 9. Latin-hypercube robustness analysis.

Matched- i analysis

Because $iota = i^*h_i/(1+i^*h_i)$, multiple pairs of i and h_i produce the same instantaneous inhibitor activity. Endpoint similarity within a matched group supports treating $iota$ as the direct pathway variable, while remaining differences in free-Bla AUC, antibiotic exposure, or recovery indicate information carried by other rates or by numerical/transient effects. Varying d_b at fixed $iota$ provides a direct test of the free-Bla time scale.



Figures 10-11. Matched inhibitor activity and fixed-activity d_b identifiability checks.

Interpretation, implications, and limitations

1. Inhibitor activity has two linked ecological effects in the full model: beta moves toward one, weakening private lysis protection, and phi decreases, weakening resistant-mediated antibiotic degradation. A dose setting can therefore change both selection and the depth/duration

of the population bottleneck. 2. h_i is not a monotonic 'strength' parameter over the entire dose range. Below $i=1$ it lowers $iota$ relative to a lower h_i , while above $i=1$ it raises $iota$; the response atlas and pathway plot should be interpreted with this sign change in mind. 3. d_b is a time-scale parameter. It can leave the instantaneous inhibitor activity unchanged while changing free-Bla persistence, antibiotic exposure, recovery timing, and possibly the final composition. 4. Initial composition is not a nuisance detail. The same dynamic selection effect can produce different final f_R values depending on $f_R(0)$, so Δf_R and $\Delta \log(n_r/n_s)$ should be reported with any endpoint claim. 5. All conclusions remain conditional on the dimensionless base model, no spatial structure, no horizontal gene transfer, constant treatment parameters, and the group's choices for antibiotic anchors and endpoint thresholds.

Hand-off to the other roles

Person 1 should confirm α , $\beta_{\min}$, c , the selection criterion, and the final robustness ranges. Person 2 should confirm the a_0 , γ , and h_a anchors and whether the high-dose horizon is appropriate. Person 4 should confirm κ_b , d_a , $\phi_{\max}$, ξ uncertainty, direct a/b observables, and sampling timing. The group should then agree on the state order, solver tolerances, extinction/recovery thresholds, and which coordination runs belong in the final shared report. Until those decisions are made, the tables and figures in this report should be treated as a reproducible local analysis package rather than final group claims.

Comprehensive inhibitor analysis: full parameter sweep

To match the breadth of the other workstreams, this addendum evaluates a structured $21 \times 21 \times 21$ grid (9,261 simulations) over inhibitor dose i , inhibitor Hill coefficient h_i , and free-Bla inactivation rate d_β , at the shared Ma-style antibiotic anchor $a_0=2.15$. It reports the same family of endpoint maps, marginal effects, sensitivity models, tradeoffs, trajectory contrasts, and outcome regimes used in the other notebooks. The previously generated multi-anchor response atlas, composition analysis, robustness runs, identifiability study, ablations, and heatmaps remain included above.

Endpoint	Minimum	Median	Maximum
Change in resistant fraction	-0.09275	-0.009025	0.008507
Final resistant fraction	0.4072	0.491	0.5085
Final Shannon diversity	0.6758	0.693	0.6931
Minimum Shannon diversity	0.6758	0.6913	0.6928
Final total population	3.636	4.01	4.188
Time to steady state	9	12	22
Sensitive recovery time	7	10	20
Peak total population loss	0.1423	0.2693	0.569
Final susceptible population	2.018	2.068	2.178
Final resistant population	1.481	1.966	2.12

Outcome classification over the full joint grid: community extinct: 0 (0.0%); nonconverged near-extinction: 0 (0.0%); nonconverged viable density: 0 (0.0%); susceptible-dominated: 0 (0.0%); resistant-dominated: 0 (0.0%); mixed community: 9,261 (100.0%). Dominance uses $f_R \leq 0.25$ or ≥ 0.75 ; viability uses 50% of the no-treatment susceptible endpoint; extinction is reported at 10^{-4} for individual strains and 10^{-6} for the whole community. Across this joint grid, the median Δf_R is -0.0090, with range [-0.0928, 0.0085]; selection-direction fractions are susceptible-enriched: 55.4%, resistant-enriched: 42.7%, neutral: 1.9%.

Validation and selection criterion

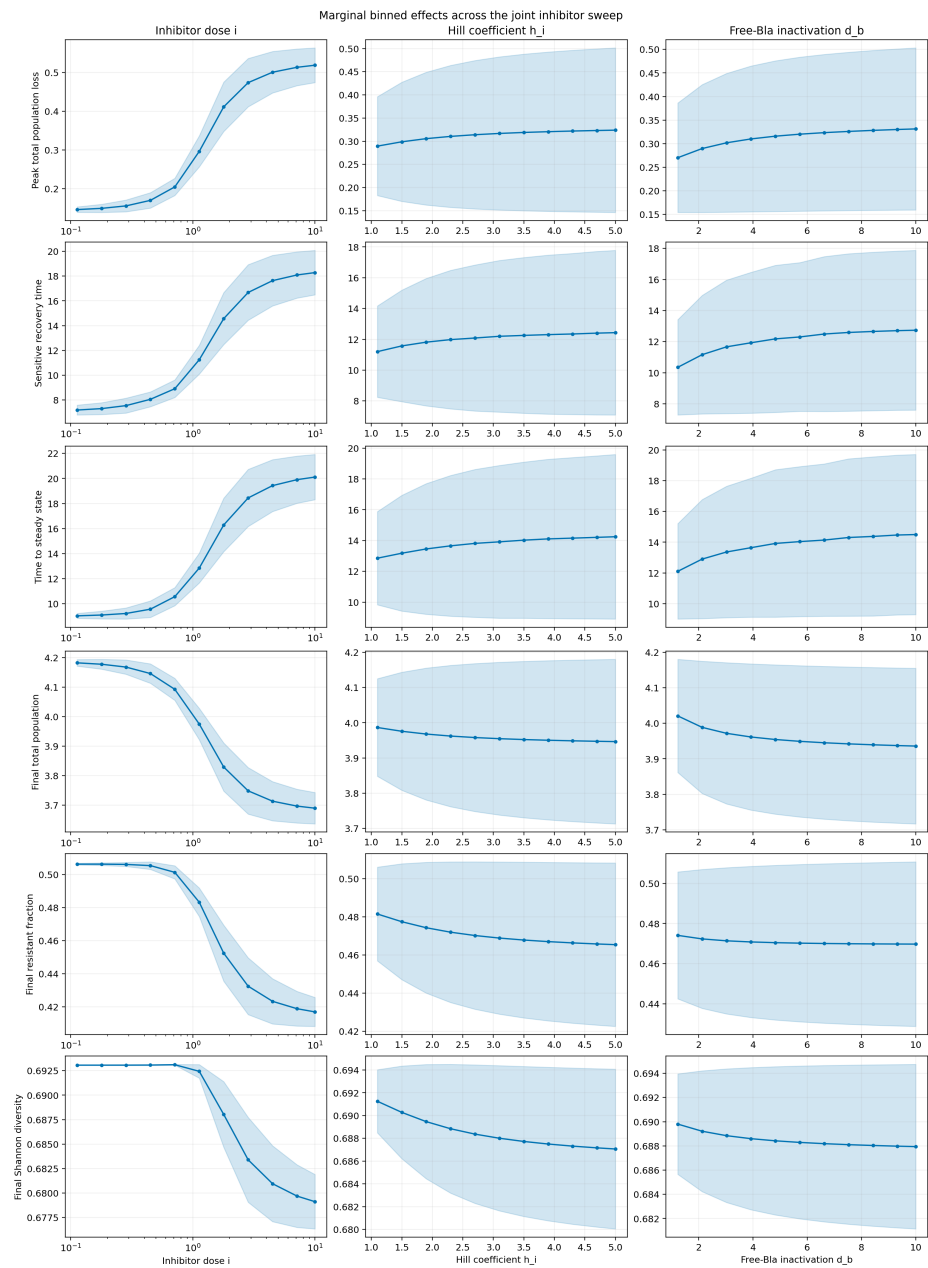
The log-population solver was compared with direct linear-state integration at 12 randomized (i , h_i , d_β) settings. All solver integrations succeeded; 12/12 checks met the combined absolute/relative tolerance (maximum normalized error 0.631). Equivalence status: PASS. The time-varying simulations are compared with the paper's instantaneous selection criterion evaluated at the initial nutrient and antibiotic state. Across 9,261 inhibitor combinations, that initial-state criterion agrees with the sign of the full-trajectory log-ratio change in 78.7% of runs. Mismatches are expected because i , h_i , and d_β interact with the evolving antibiotic/Bla environment; d_β changes the trajectory but is absent from the instantaneous criterion. No-treatment-relative susceptible viability was met in 100.0% of runs; resistant extinction occurred in 0.0%; and 100.0% met the shared steady-state detector by $\tau=300$.

Global sensitivity and marginal response

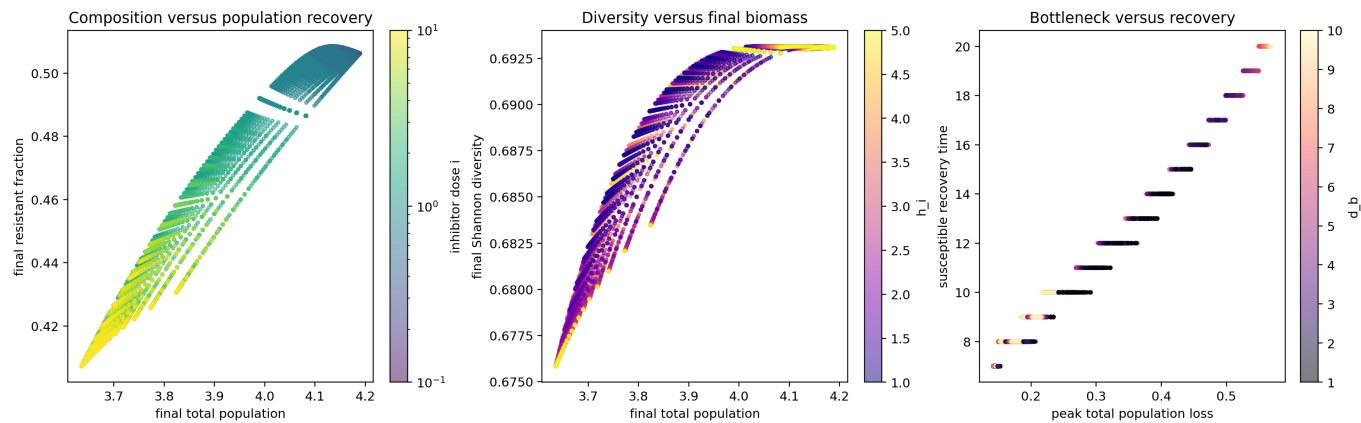
The standardized regression includes standardized main effects and all pairwise interactions among i , h_i , and d_β . Coefficients are conditional associations over this structured grid, not causal effects outside the stated model. Binned curves show the mean and ± 1 standard deviation across the other two varied parameters.

Endpoint	Strongest main β	Strongest interaction β	Top Spearman ρ	R^2
Change in resistant fraction	i : -0.841	$i:h_i$: -0.071	i : -0.908	0.732
Final Shannon diversity	i : -0.843	$i:h_i$: -0.130	i : -0.823	0.790
Final total population	i : -0.811	$i:h_i$: -0.071	i : -0.940	0.681
Sensitive recovery time	i : +0.813	$i:d_\beta$: +0.103	i : +0.932	0.705
Peak total population loss	i : +0.812	$i:d_\beta$: +0.079	i : +0.940	0.685

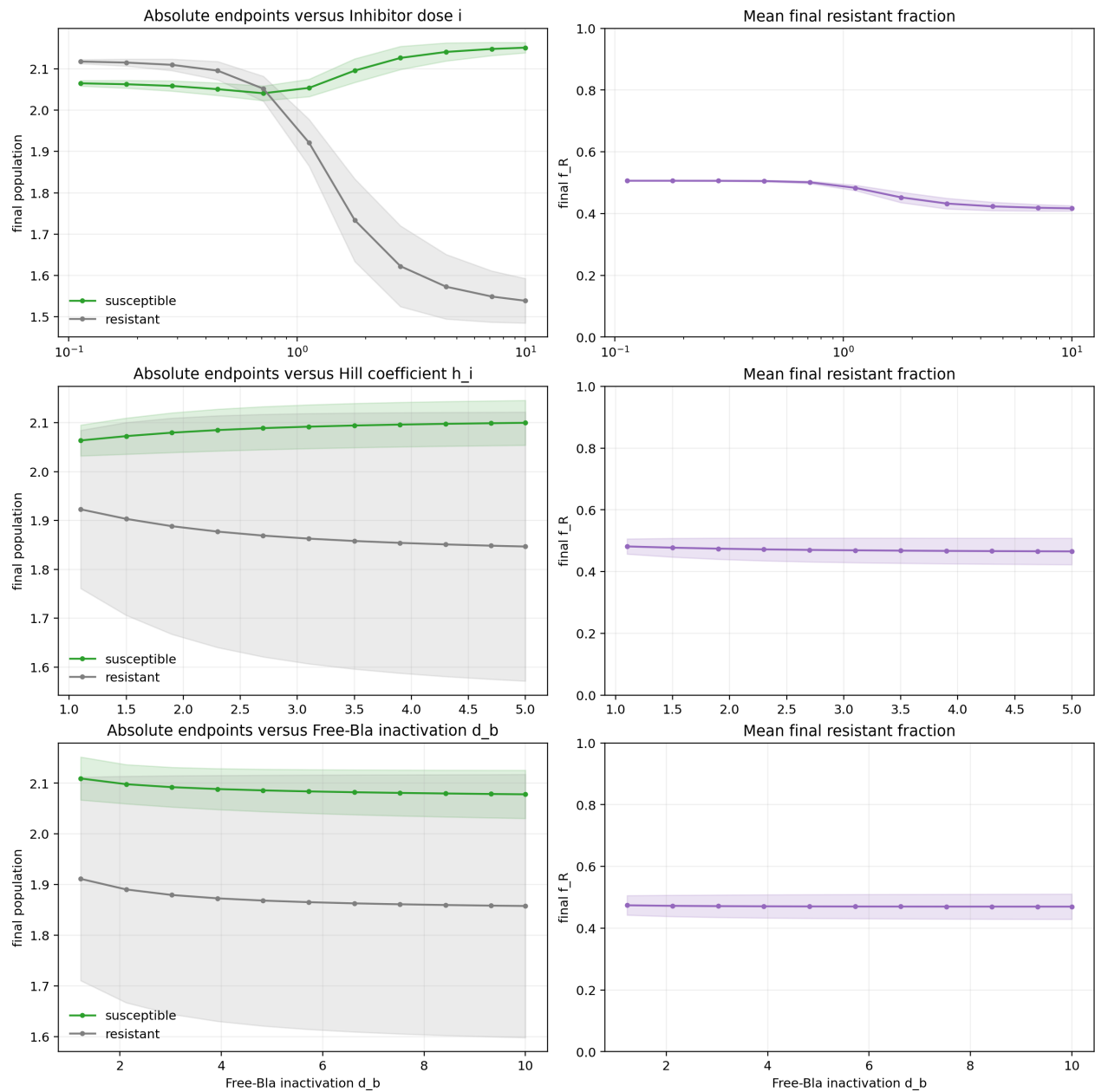
Marginal binned endpoint responses



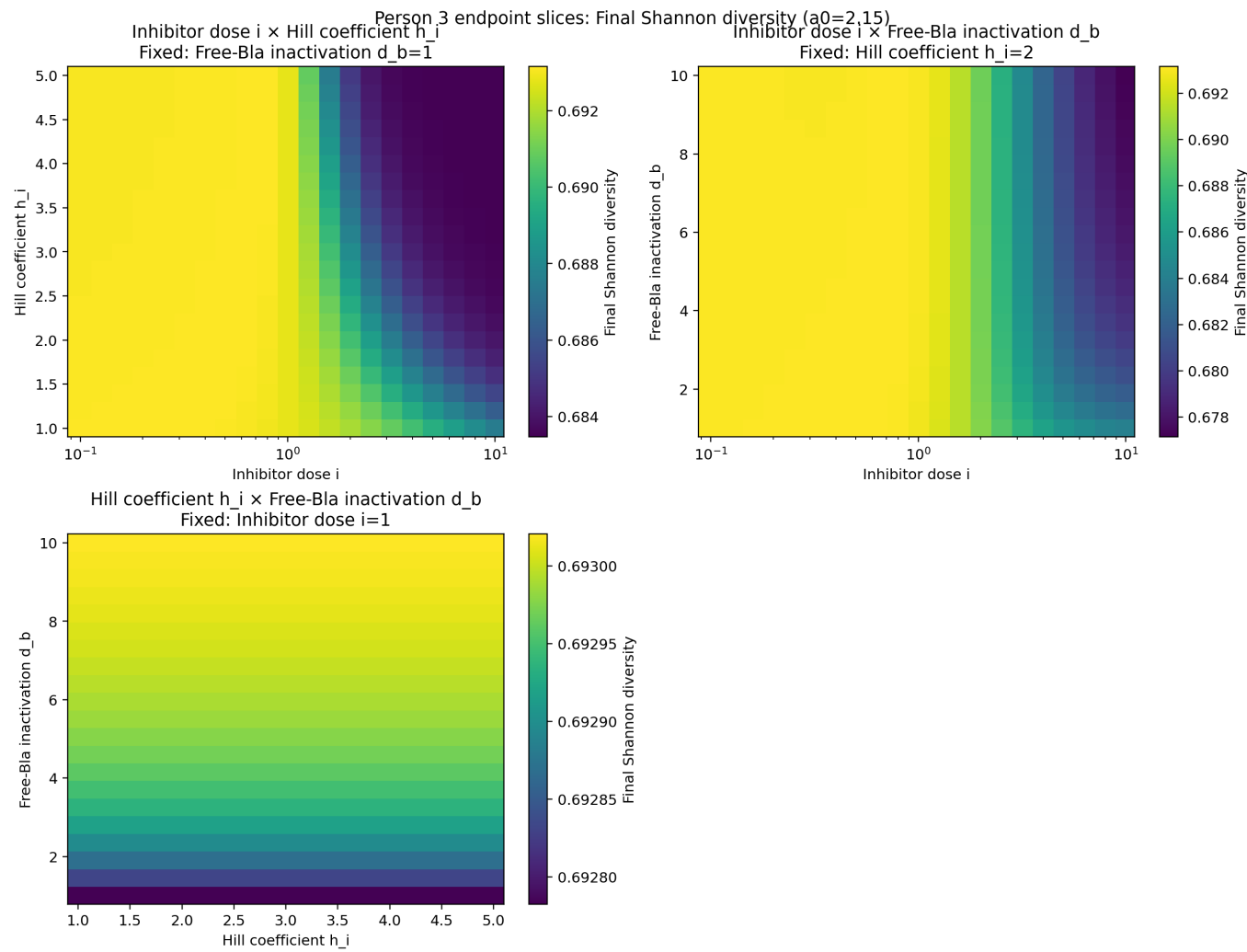
Endpoint tradeoffs: composition, diversity, bottleneck, and recovery



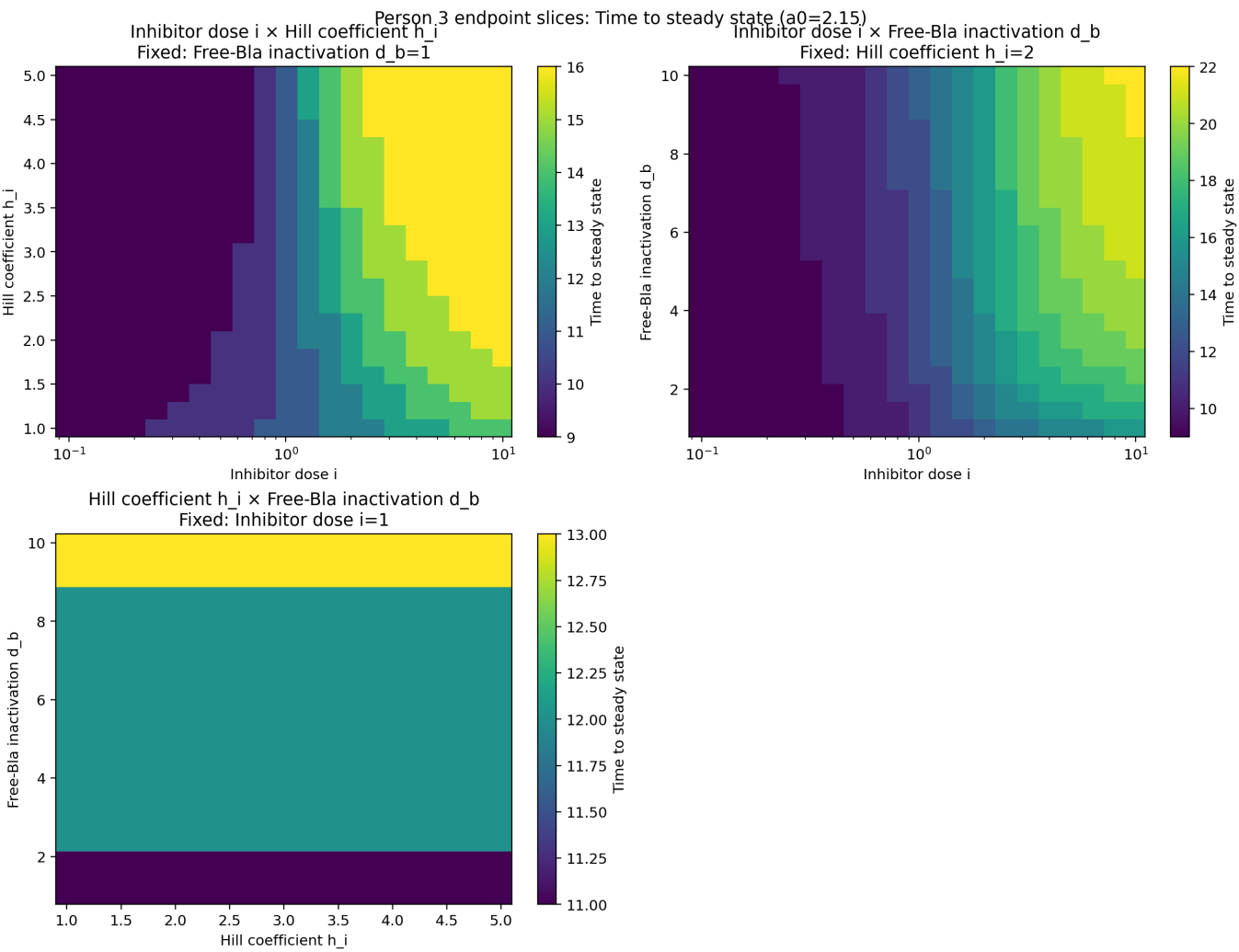
Absolute population and final composition versus inhibitor parameters



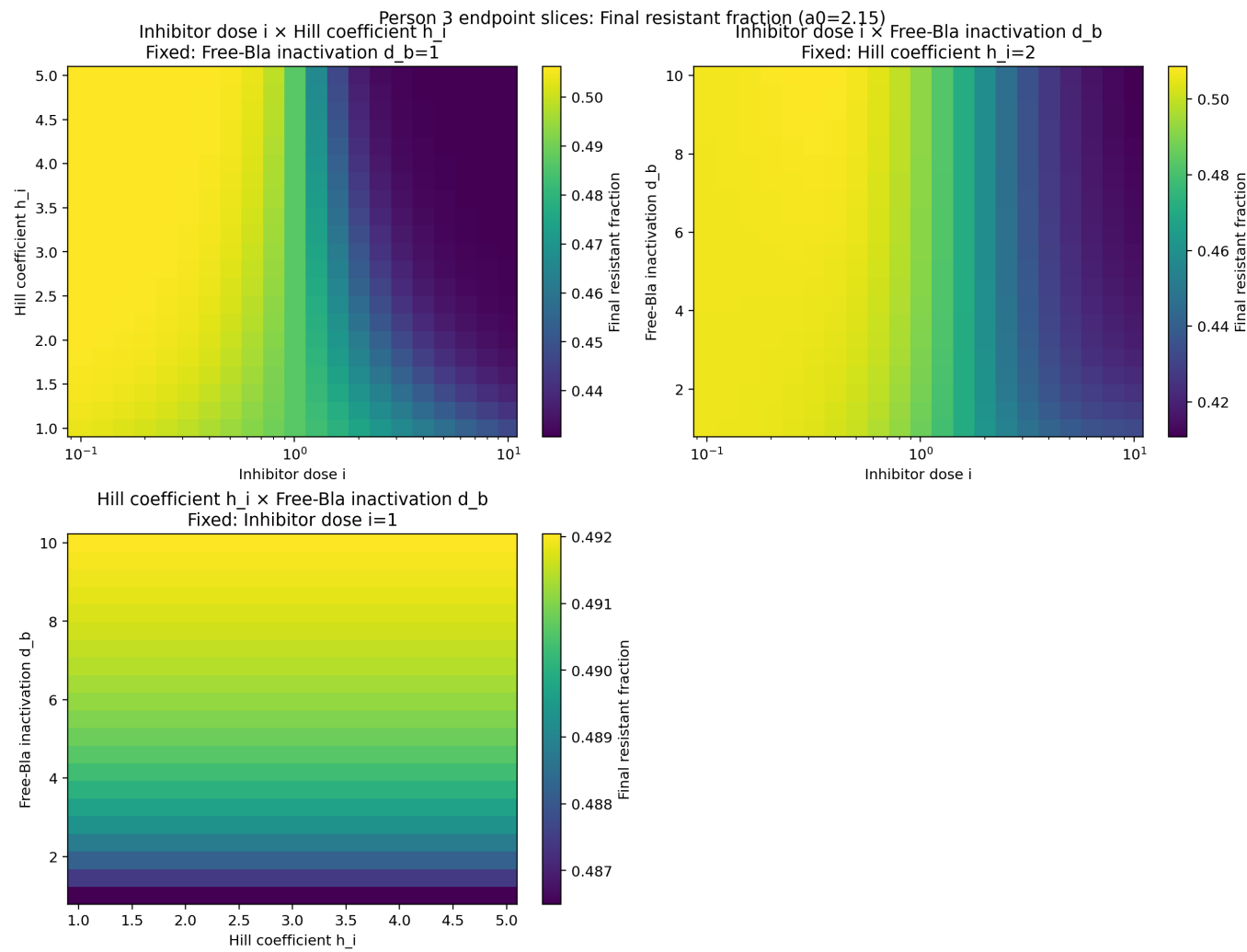
Pairwise heatmaps: Final Shannon diversity



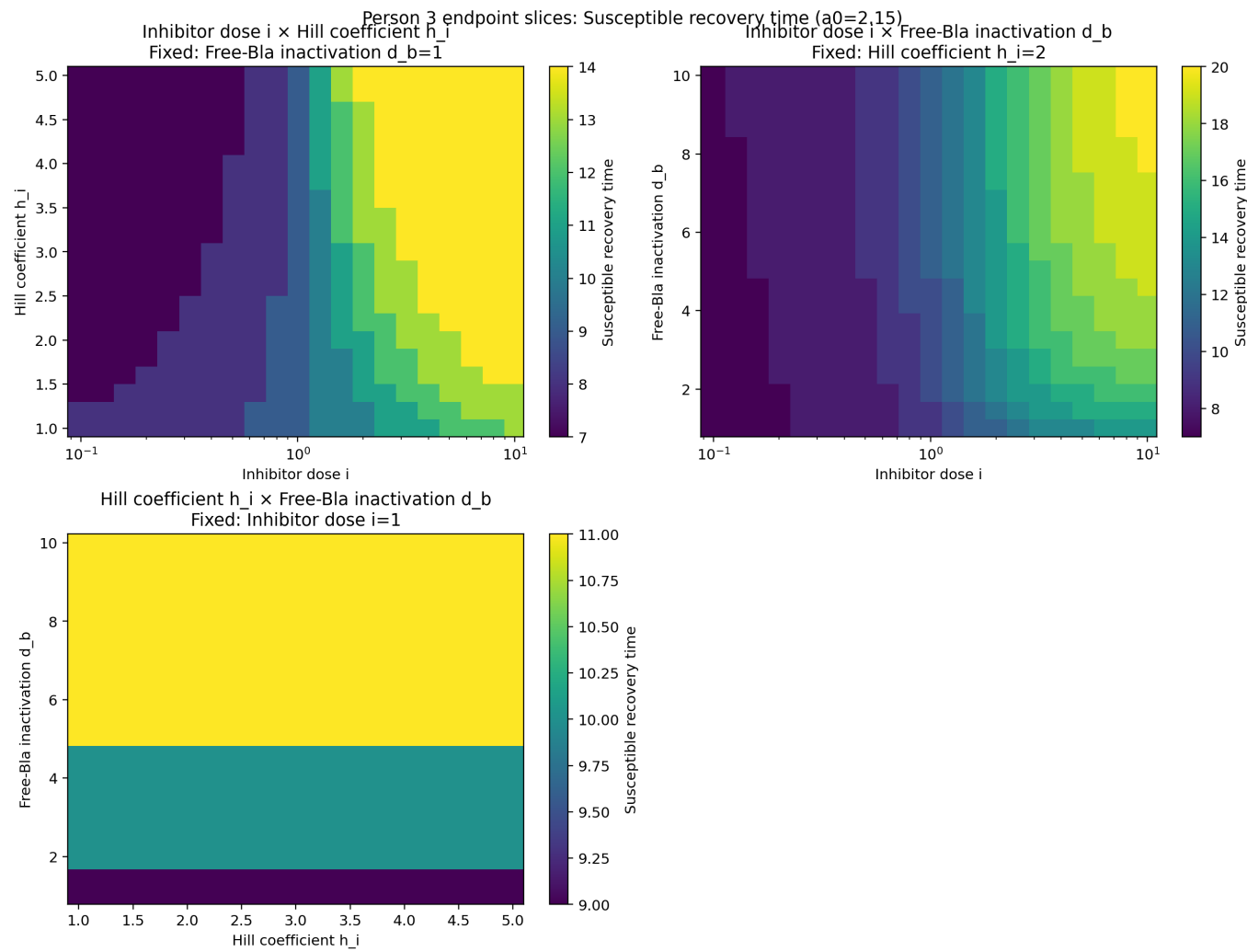
Pairwise heatmaps: Time to steady state



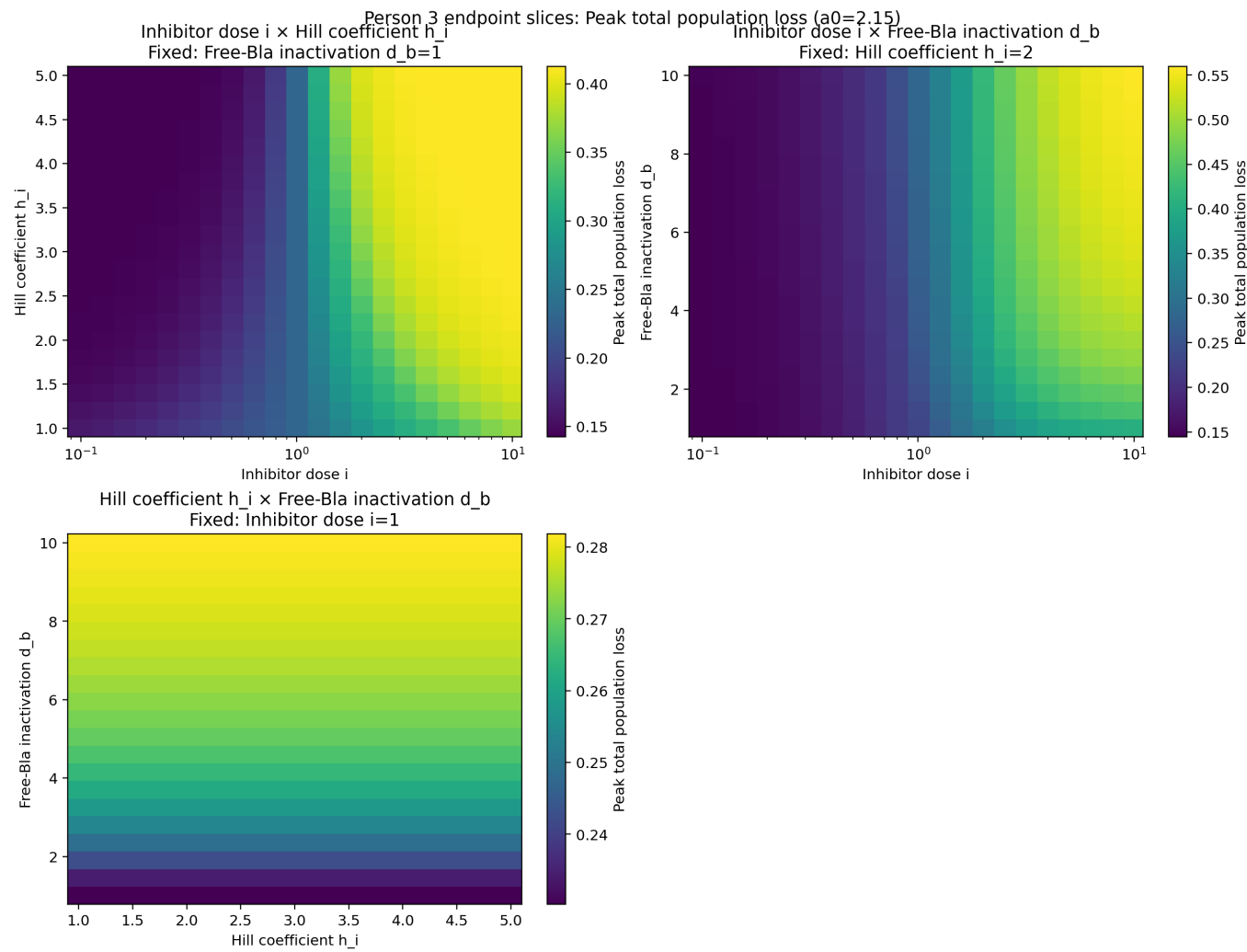
Pairwise heatmaps: Final resistant fraction



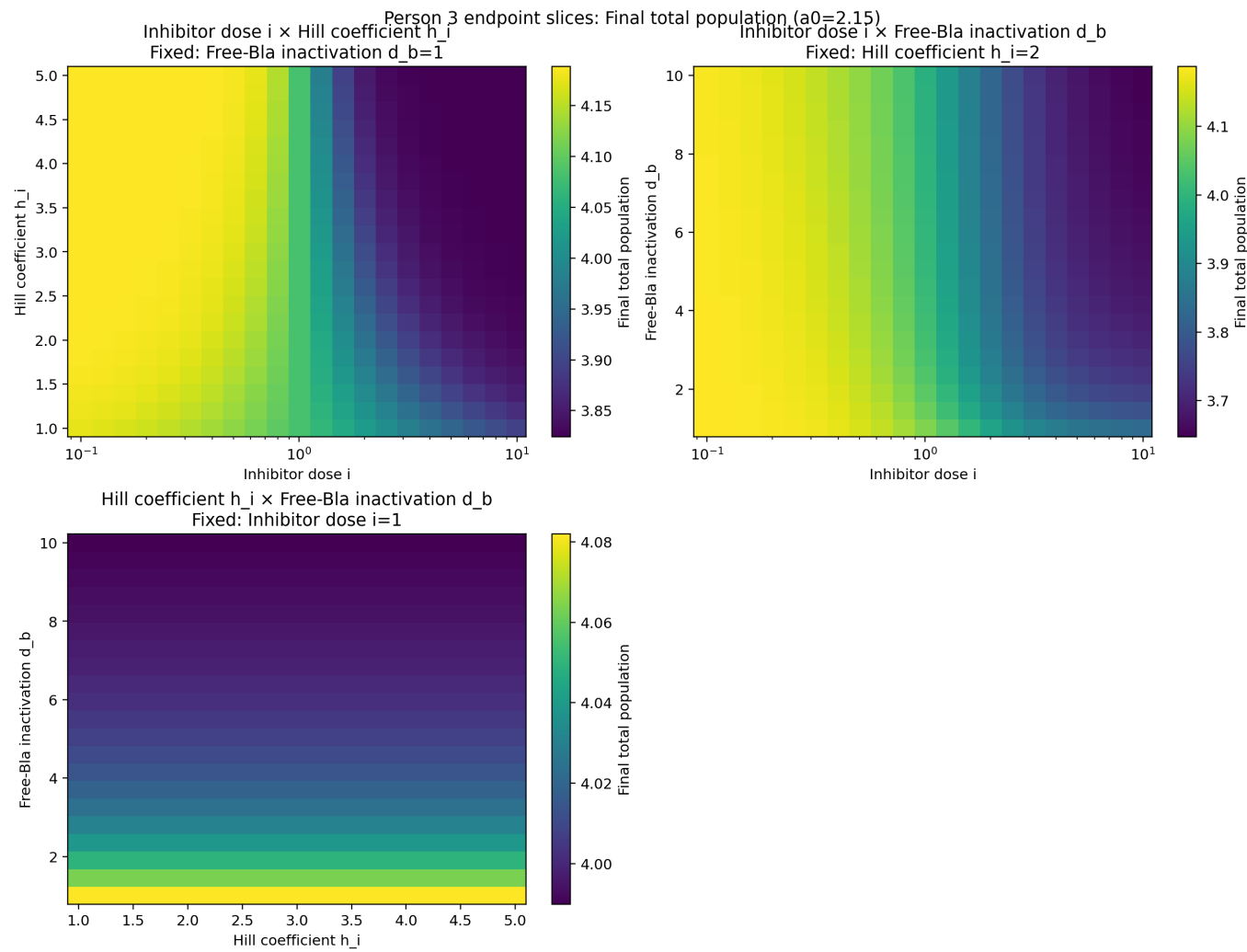
Pairwise heatmaps: Sensitive recovery time



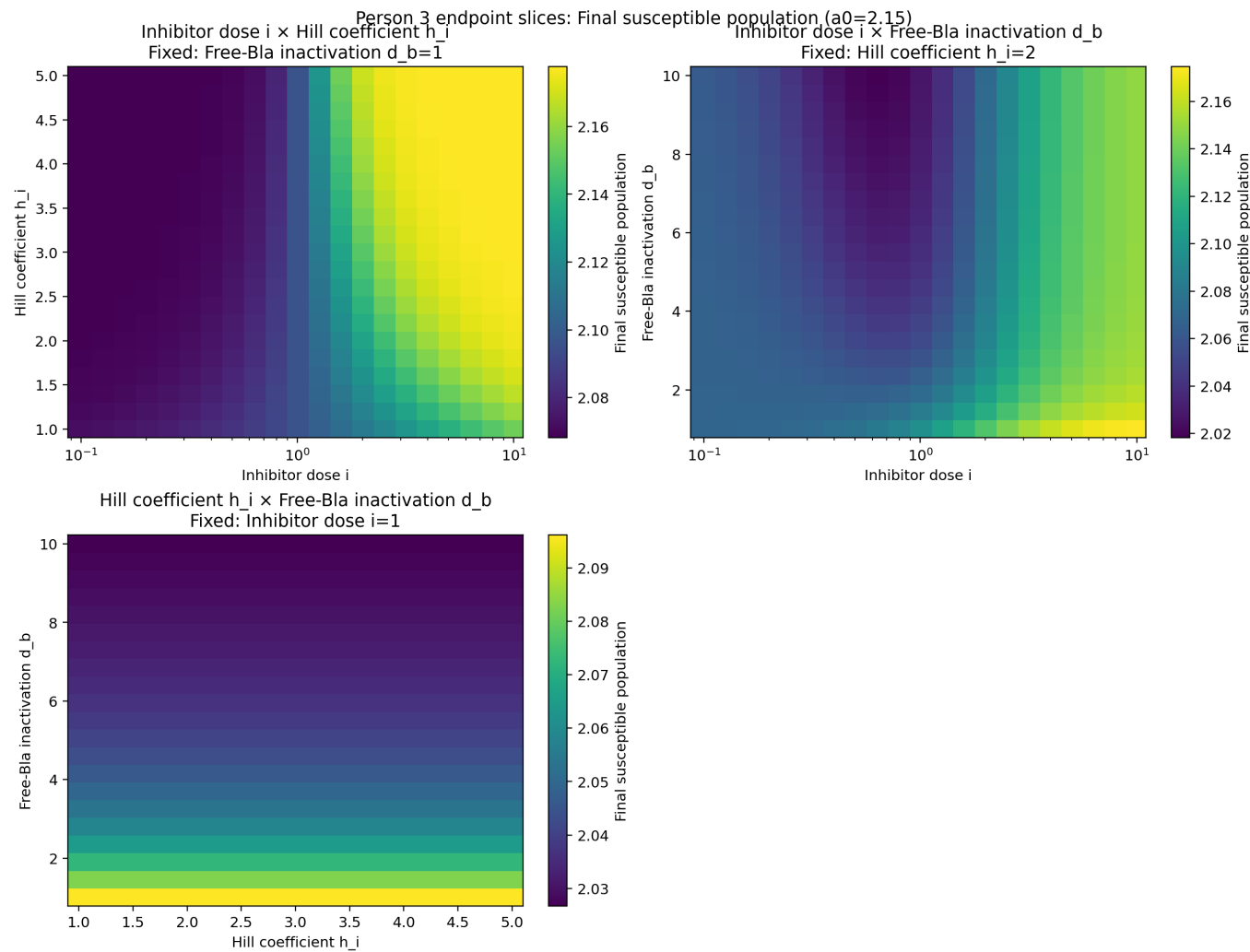
Pairwise heatmaps: Peak total population loss



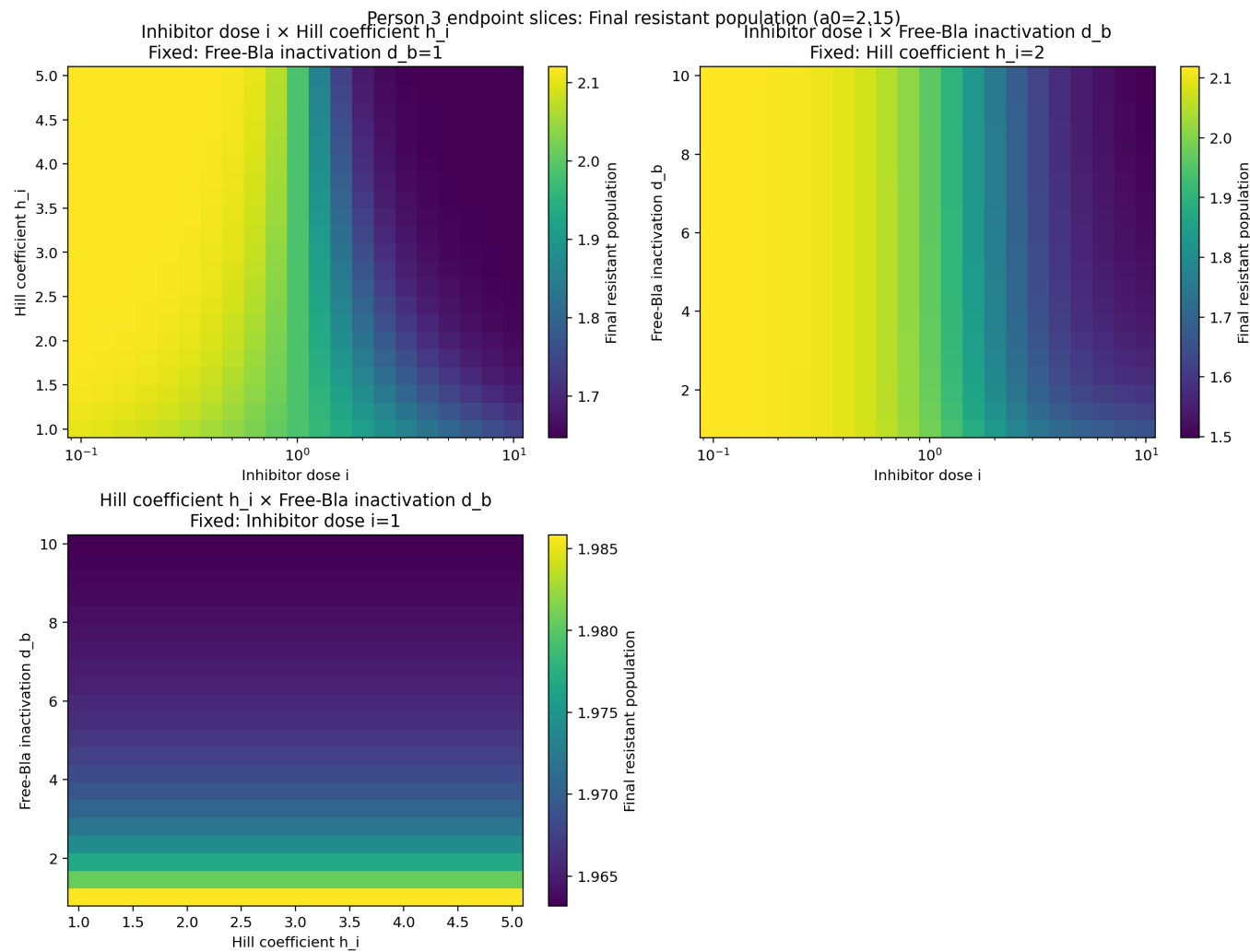
Pairwise heatmaps: Final total population



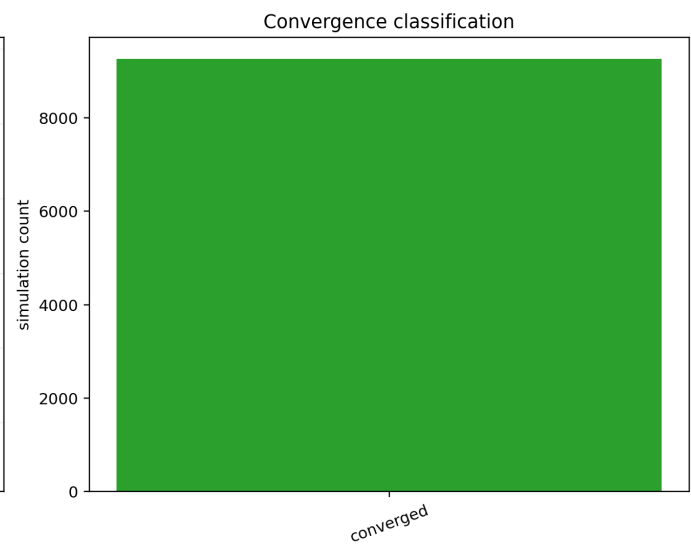
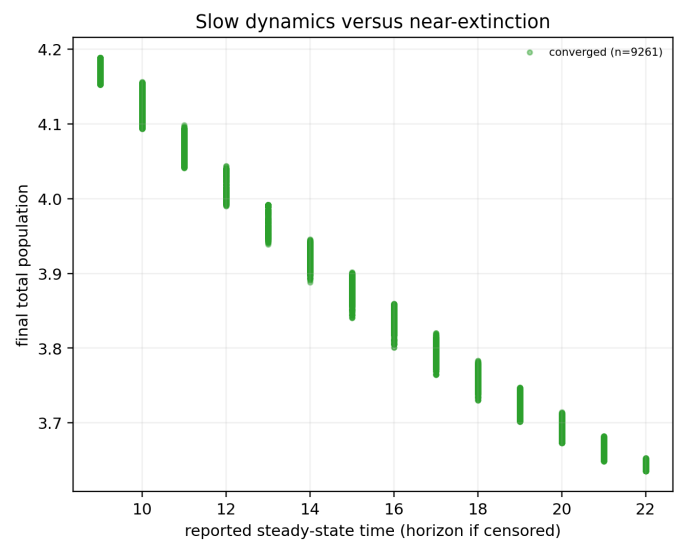
Pairwise heatmaps: Final susceptible population



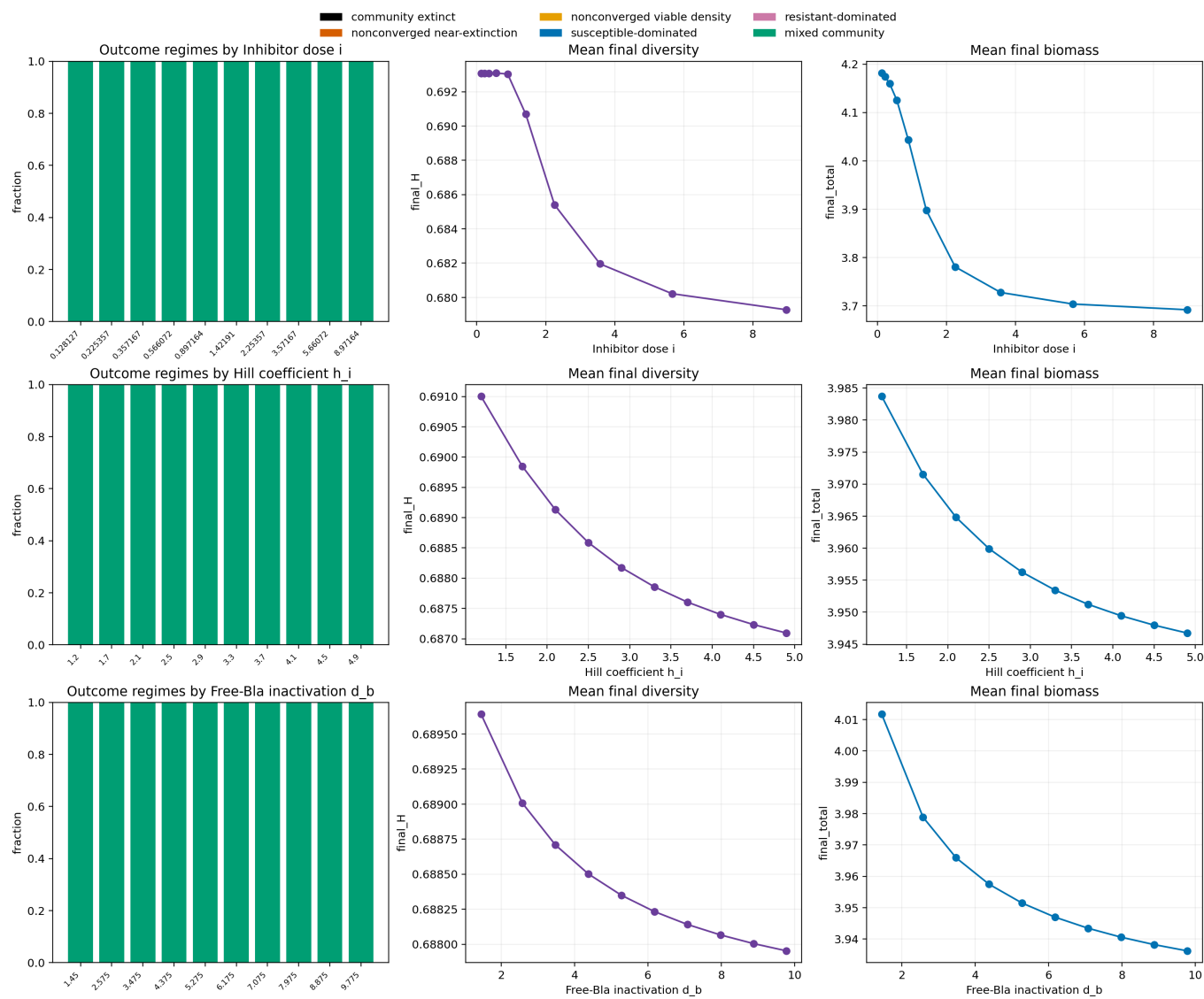
Pairwise heatmaps: Final resistant population



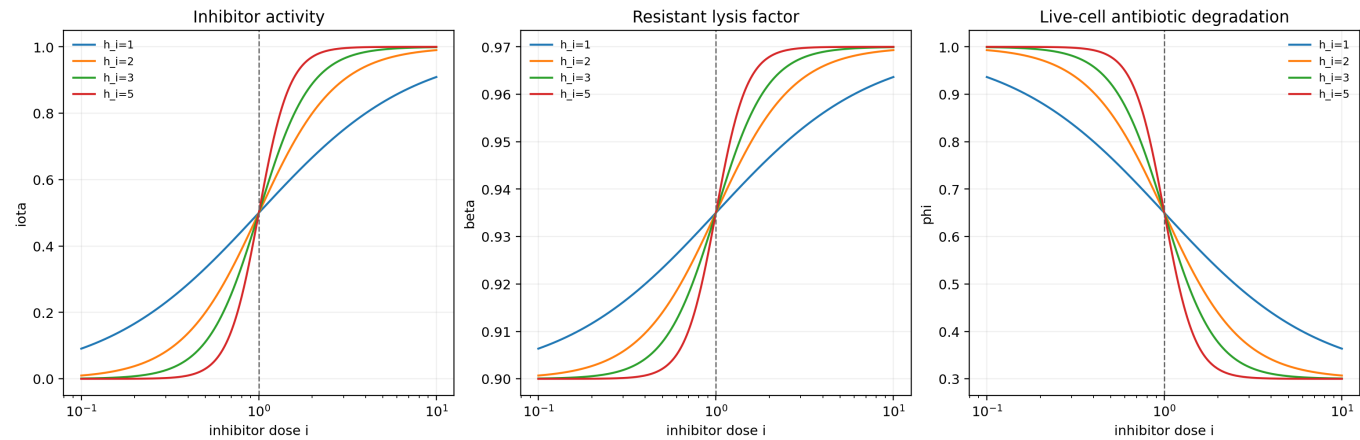
Convergence diagnostics



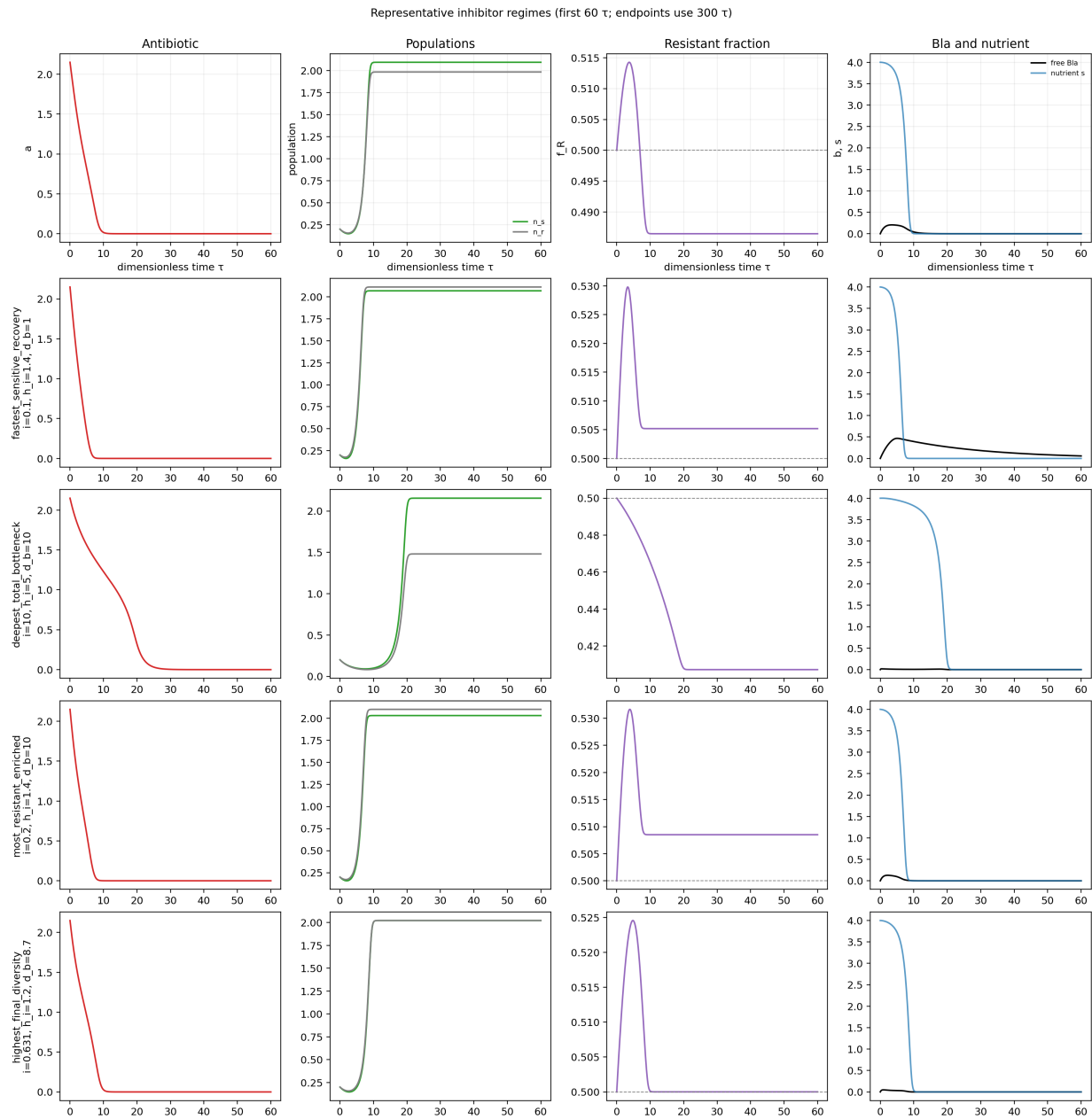
Outcome regimes, diversity, and final biomass



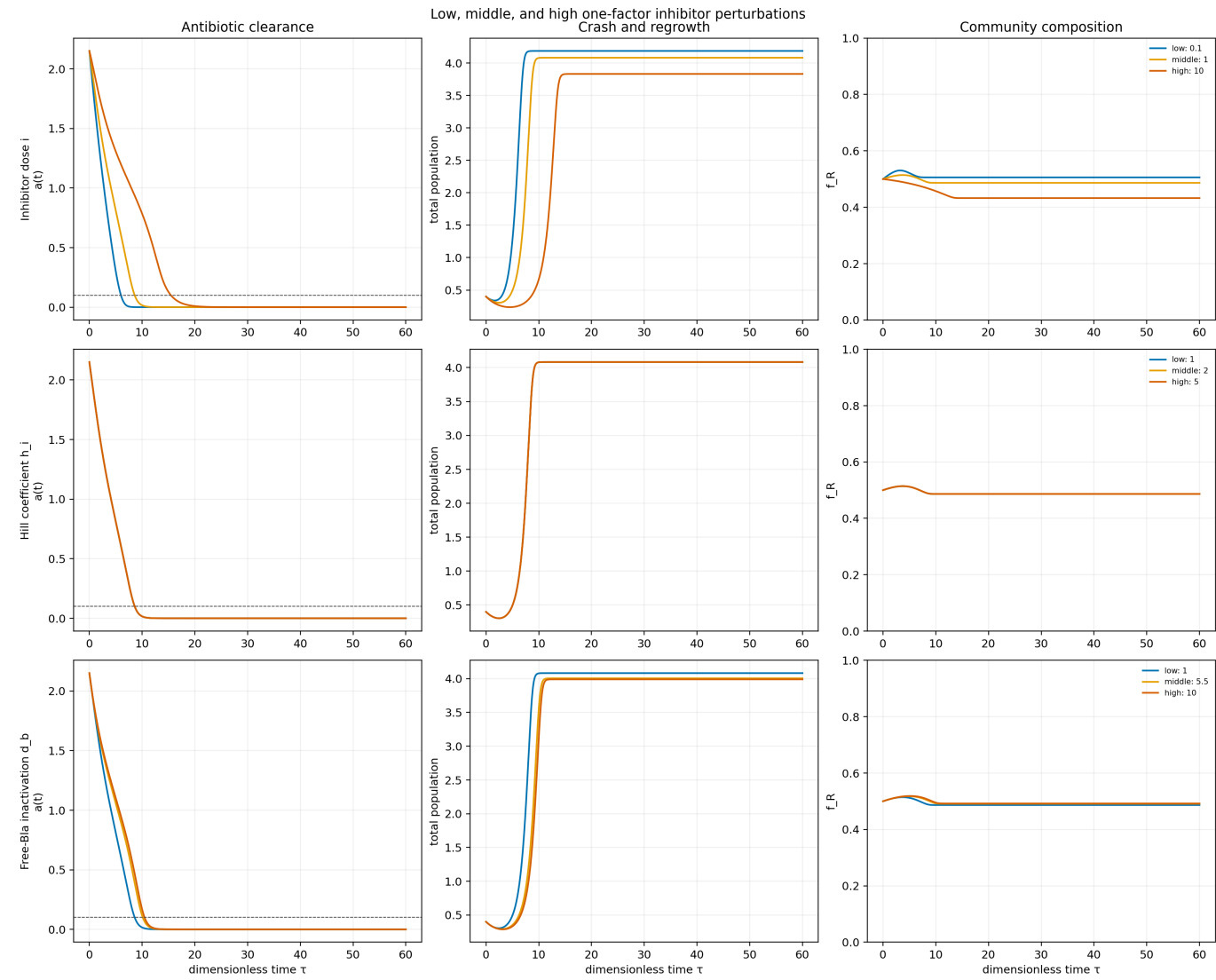
Direct inhibitor pharmacodynamic functions



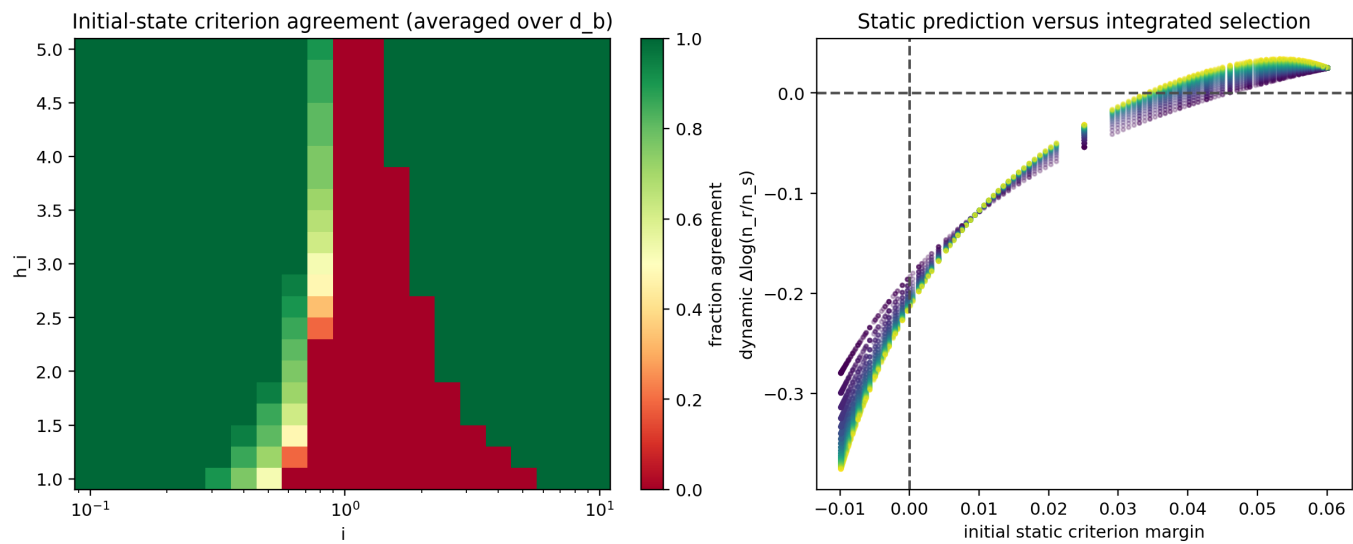
Selected full-grid representative trajectories (first 60 τ ; endpoint metrics use 300 τ)



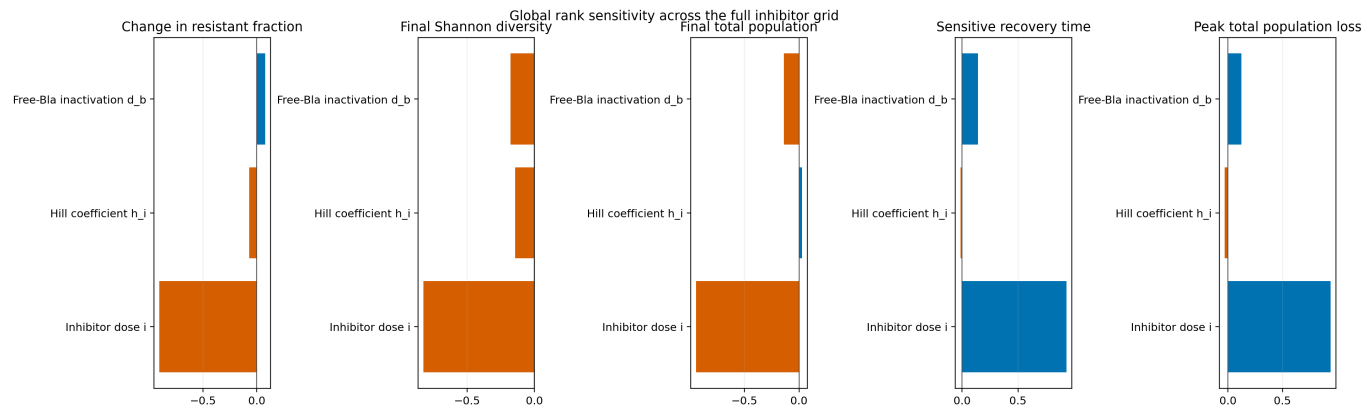
Low, middle, and high inhibitor perturbation trajectories



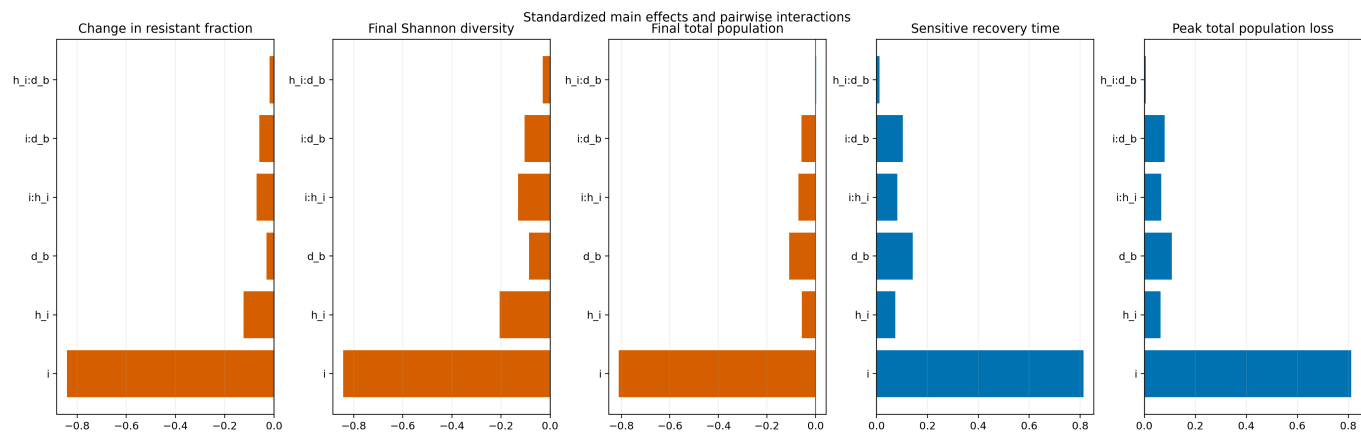
Static selection criterion compared with dynamic outcomes



Global sensitivity: 10 spearman sensitivity



Global sensitivity: 11 standardized interactions



Sensitivity tables and reproducibility

Machine-readable full-grid results, endpoint summaries, binned response summaries, Spearman associations, standardized regression coefficients, criterion comparisons, convergence classifications, and trajectory-case selections are saved in results/tables/ with the `comprehensive\_` prefix. The runner is person3_comprehensive_analysis.py. All comprehensive figures have both PNG and PDF versions under results/figures/ and are mirrored to the top-level figures/ folder.